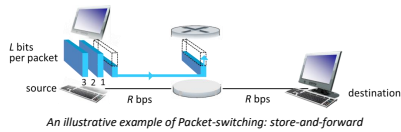


1. As we introduced in the lecture, the packet switching uses **store-and-forward transmission** at the inputs to the links. That means the entire packet must arrive at router before it can be transmitted onto the next link. (Ignore queuing, propagation delay, and processing delay)



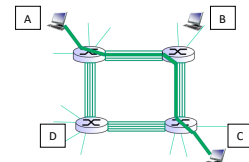
- a. Suppose there is exactly one packet switch between a sending host and a receiving host. The transmission rates between the sending host and the switch and between the switch and the receiving host are R_1 and R_2 , respectively. With store-and-forward switching, what is the total end-to-end delay to send a packet of length L ?

- A) L / R_1
 B) L / R_2
 C) $L / R_1 + L / R_2$
 D) $2L / R_1 + 2L / R_2$

- b. Assuming the transmission rate to be constant at R , what will be the end-to-end delay for sending $P=3$ packets each of length L across $N=2$ links (i.e., the packets need to pass two links to reach the destination as shown in the figure above)?

- A) L / R
 B) $2L / R$
 C) $3L / R$
 D) $4L / R$

2. Consider the circuit switch network in Figure below. Suppose that there are 4 circuits on each link. Label the four switches A, B, C and D, going in the clockwise direction.



- a. What is the maximum number of simultaneous connections that can be in progress at any one time in this network?

- A) 4
 B) 8
 C) 16
 D) 32

- b. Suppose that all connections are between switches A and C. What is the maximum number of simultaneous connections that can be in progress?

- A) 4
 B) 8
 C) 16
 D) 32

- c. Suppose we want to make four connections between switches A and C, and another four connections between switches B and D. Can we route these calls through the four links to accommodate all eight connections?

- A) No
 B) Yes

3. Suppose we want to send a file of 160,000 bits from Host A to Host B over a circuit-switched network. Suppose that all links in the network use FDM with 12 frequency bands of the same bandwidth. Assume that the 12 frequency bands together can support a total bit rate of 1.536 Mbps. Also suppose that it takes 600 msec to establish an end-to-end circuit before Host A can

begin to transmit the file. How long does it take to send the file?

- A) 1.85 sec
 B) 1.95 sec
 C) 2.05 sec
 D) 2.15 sec

4. Suppose users share a 3 Mbps link. Also suppose each user requires 150 kbps when transmitting, but each user transmits only 10 percent of the time.

- a. When circuit switching is used, how many users can be supported?

- A) 10
 B) 20
 C) 30
 D) 40

- b. For the remainder of this problem, suppose packet switching is used. Find the probability that a given user is transmitting.

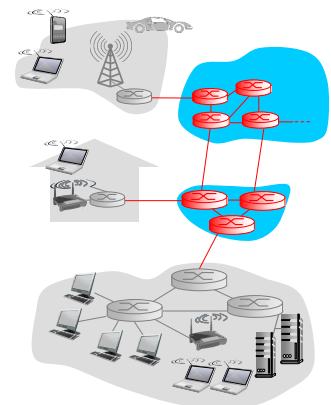
- A) 0.1
 B) 0.2
 C) 0.3
 D) 0.4

- c. Suppose there are 120 users. Find the probability that at any given time, exactly n users are transmitting simultaneously.

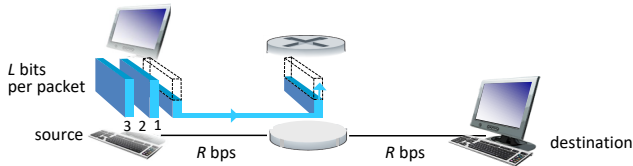
- A) $\binom{120}{n} (0.1)^{120-n} (0.9)^n$
 B) $\binom{120}{n} (0.9)^{120-n} (0.1)^n$
 C) $(0.9)^n$
 D) $(0.1)^n$

The network core

- ❖ mesh of interconnected routers
- ❖ packet-switching: hosts break application-layer messages into packets
 - forward packets from one router to the next, across links on path from source to destination
 - each packet transmitted at full link capacity



Packet-switching: store-and-forward



- ❖ takes L/R seconds to transmit (push out) L -bit packet into link at R bps
 - ❖ **store and forward**: entire packet must arrive at router before it can be transmitted on next link
 - ❖ end-end delay = $2L/R$ (assuming zero propagation delay)
- one-hop numerical example:*
- $L = 7.5$ Mbits
 - $R = 1.5$ Mbps
 - one-hop transmission delay = 5 sec
- } more on delay shortly ...

Introduction 1-2

Q1

- a) At time t_0 the sending host begins to transmit. At time $t_1 = L/R_1$, the sending host completes transmission and the entire packet is received at the router (no propagation delay). Because the router has the entire packet at time t_1 , it can begin to transmit the packet to the receiving host at time t_1 . At time $t_2 = t_1 + L/R_2$, the router completes transmission and the entire packet is received at the receiving host (again, no propagation delay). Thus, the end-to-end delay is $L/R_1 + L/R_2$.

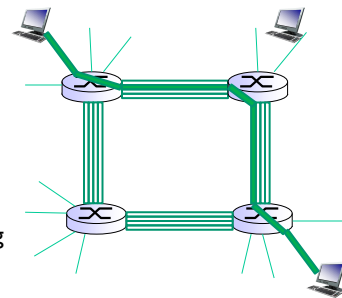
Q1

- b) Case of 2 links: When the first node sends the P -th packet to the second node, thereby incurring a net delay of PL/R , the second node sends the $(P-1)$ -th packet to the third node. Hence an additional delay of L/R would be incurred for sending the P -th packet from the second node to the third node. Thus, the net delay would be $(P+1)L/R$.

Alternative core: circuit switching

end-end resources allocated to, reserved for "call" between source & dest:

- ❖ In diagram, each link has four circuits.
 - call gets 2nd circuit in top link and 1st circuit in right link.
- ❖ dedicated resources: no sharing
 - circuit-like (guaranteed) performance
- ❖ circuit segment idle if not used by call (*no sharing*)
- ❖ Commonly used in traditional telephone networks



Introduction 1-5

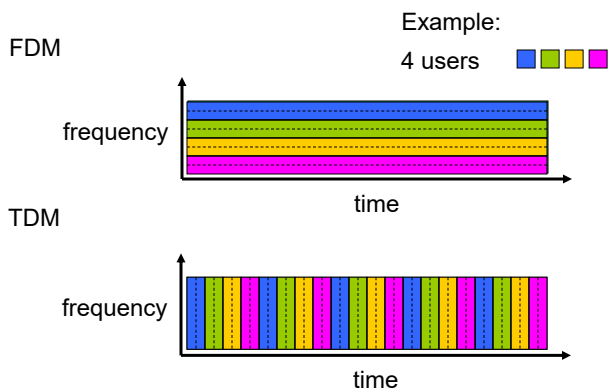
Q2

- a) Between the switch in the upper left and the switch in the upper right we can have 4 connections. Similarly we can have four connections between each of the 3 other pairs of adjacent switches. Thus, this network can support up to 16 connections.
- b) We can 4 connections passing through the switch in the upper-right-hand corner and another 4 connections passing through the switch in the lower-left-hand corner, giving a total of 8 connections.

Q2

- C) Yes. For the connections between A and C, we route two connections through B and two connections through D. For the connections between B and D, we route two connections through A and two connections through C. In this manner, there are at most 4 connections passing through any link.

Circuit switching: FDM versus TDM



Introduction 1-8

Q3

- Each circuit has a transmission rate of $(1.536 \text{ Mbps})/12 = 128 \text{ kbps}$, so it takes $(160,000 \text{ bits})/(128 \text{ kbps}) = 1.25 \text{ seconds}$ to transmit the file. To this, we add the circuit establishment time of 0.6 seconds, giving a total of 1.85 seconds to send the file.

Q4

- a) 3Mbps/150 kbps = 20 users can be supported.
- b) $p = 0.1$
- c) $\binom{120}{n} p^n (1-p)^{120-n}$
- Binomial distribution $B(N, p)$: assume the success probability of an independent trial is p , then the probability of getting exactly n successes in N trials is $\binom{N}{n} p^n (1-p)^{N-n}$